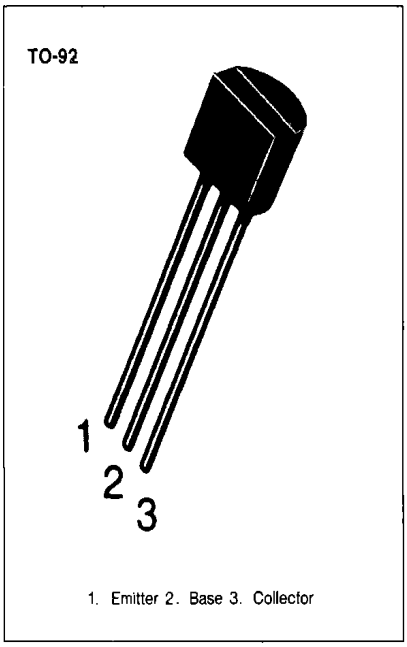
Transistors
2SA642

USHA (INDIA) LTD

LOW FREQUENCY POWER AMPLIFIER

- Collector Dissipation P_C=400mW



1. Emitter 2. Base 3. Collector

ABSOLUTE MAXIMUM RATINGS (T_a=25°C)

Characteristic	Symbol	Rating	Unit
Collector-Base Voltage	V_{CBO}	−30	V
Collector-Emitter Voltage	V_{CEO}	−25	V
Emitter-Base Voltage	V_{EBO}	−5	V
Collector Current (DC)	I_C (DC)	−300	mA
Collector Current (pulse)	I_C(pulse)*	−500	mA
Collector Dissipation	P_C	400	mW
Junction Temperature	Tj	150	°C
Storage Temperature	Tstg	−55~150	°C

* PW≤10ms, duty cycle ≤50%

ELECTRICAL CHARACTERISTICS (T_a=25°C)

Characteristic	Symbol	Test Conditions	Min	Typ	Max	Unit
Collector-Base Breakdown Voltage	BV_{CBO}	I_C=−100μA, I_E=0	−30			V
Collector-Emitter Breakdown Voltage	BV_{CEO}	I_C=−10mA, I_B=0	−25			V
Emitter-Base Breakdown Voltage	BV_{EBO}	I_E=−10μA, I_C=0	−5			V
Collector Cut-off Current	I_{CBO}	V_{CB}=−25V, I_E=0			−100	nA
Emitter Cut-off Current	I_{EBO}	V_{EB}=−3V, I_C=0			−100	nA
DC Current Gain	h_{FE}	V_{CE}=−1V, I_C=−50mA*	70		400	
Collector-Emitter Saturation Voltage	V_{CE}(sat)	I_C=−300mA, I_B=−30mA*		−0.35	−0.6	V

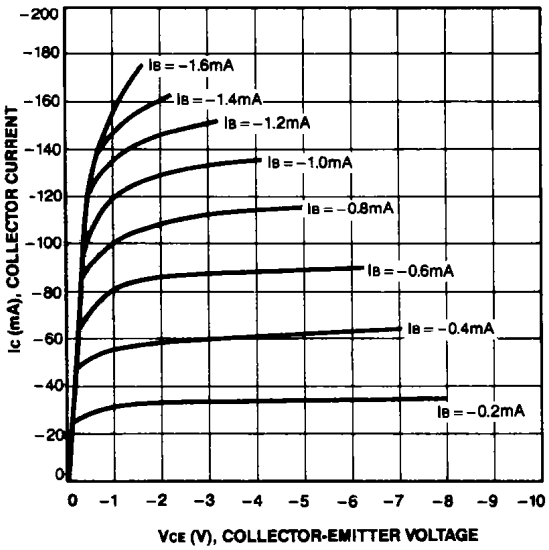
* Pulse Test: PW≤350μs, duty cycle≤2%

h_{FE} CLASSIFICATION

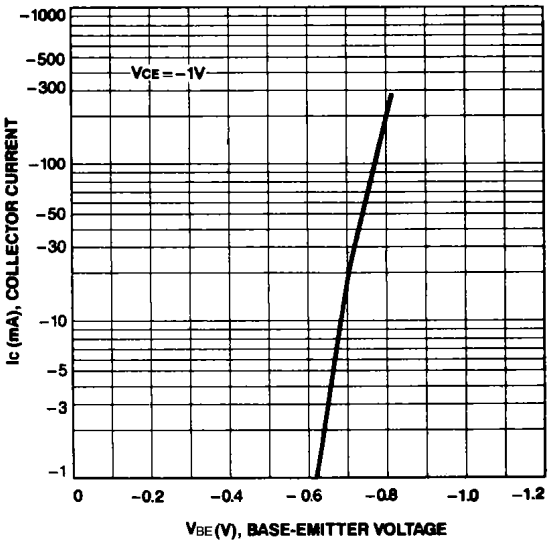
Classification	O	Y	G
h_{FE}	70-140	120-240	200-400

ISO 14001 CERTIFIED
AN ISO 9002 COMPANY

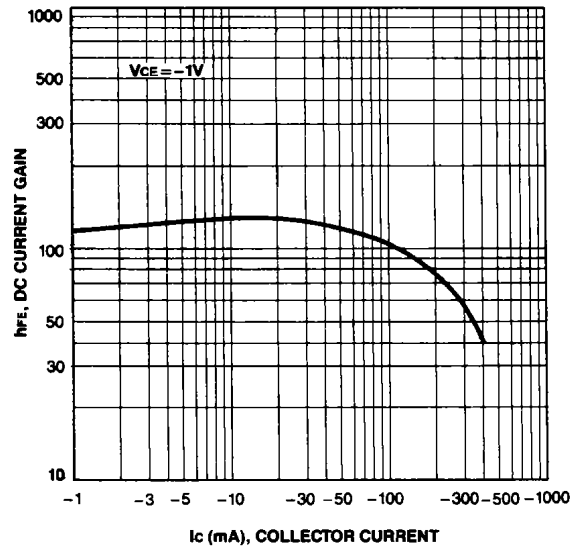
STATIC CHARACTERISTIC



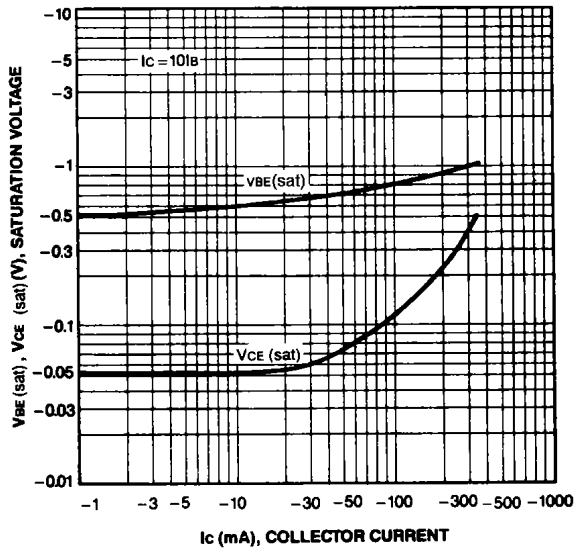
BASE-EMITTER ON VOLTAGE



DC CURRENT GAIN



BASE-EMITTER SATURATION VOLTAGE
COLLECTOR-EMITTER SATURATION VOLTAGE



COLLECTOR OUTPUT CAPACITANCE

